

The Unique Level-11 Maximizer for Finite Gilbreath Extension Sets

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Abstract

For a strictly increasing Gilbreath sequence $S = (s_1, \dots, s_n)$ beginning with $(2, 3)$, let K_S be the set of all integers that may be appended while preserving the Gilbreath property; the appended integer is not required to preserve strict increase. We give an exact computer-assisted proof that, among the 17,535,396 such sequences of length 11, the largest extension set has cardinality 1025, and that the unique equality case is $(2, 3, 5, 9, 17, 33, 65, 129, 257, 513, 1025)$. The proof combines a reverse absolute-difference recursion, an inductively complete traversal, exact integer arithmetic, and a separate full-depth implementation that reproduces the enumeration. This settles only the finite level $n = 11$; the corresponding assertion for arbitrary n is not addressed.

1 Statement and context

For an integer sequence $S = (s_1, \dots, s_n)$, set

$$a_i^{(0)} = s_i, \quad a_i^{(j)} = \left| a_{i+1}^{(j-1)} - a_i^{(j-1)} \right| \quad (1 \leq j < n, \ 1 \leq i \leq n - j).$$

The sequence is a *Gilbreath sequence* if $a_1^{(j)} = 1$ for every $1 \leq j < n$. Let $\mathcal{G}_n$ consist of the strictly increasing integer Gilbreath sequences of length n whose first two entries are 2, 3, and put

$$K_S = \{k \in \mathbb{Z} : (s_1, \dots, s_n, k) \text{ is a Gilbreath sequence}\}.$$

Strict increase is imposed on $S \in \mathcal{G}_n$, but not on the appended sequence used to define K_S .

Muney introduced this two-sided extension-set formulation and proved a reverse-preimage characterization of K_S [1, Proposition 18]. For

$$U_n = (2, 3, 5, 9, \dots, 2^{n-1} + 1),$$

Muney proved $|K_{U_n}| = 2^{n-1} + 1$ and conjectured that U_n is the unique maximizer in $\mathcal{G}_n$ [1, Theorem 29 and Conjecture 30]. The enumeration in that paper gives the complete extremal data through $n = 10$ and the count $|\mathcal{G}_{11}| = 17,535,396$, while leaving the level-11 extremal fields uncomputed. The result below supplies precisely that finite equality classification. It makes no assertion about the arbitrary- n conjecture.

Theorem 1. *For the definitions above,*

$$\max_{S \in \mathcal{G}_{11}} |K_S| = 1025.$$

Equality holds if and only if

$$S = U_{11} = (2, 3, 5, 9, 17, 33, 65, 129, 257, 513, 1025).$$

2 Exact reverse recursion

For $S \in \mathcal{G}_n$, let

$$c_j = a_{n-j}^{(j)} \quad (0 \leq j < n)$$

be the rightmost entry in row j of its absolute-difference triangle; in particular, $c_0 = s_n$. Define a preimage operator on finite subsets of $\mathbb{Z}_{\geq 0}$ by

$$P_c(Y) = \{c + y : y \in Y\} \cup \{c - y : y \in Y, y \leq c\}.$$

Lemma 2 (reverse-edge formula). *Set $Y_n = \{1\}$ and, successively for $j = n - 1, n - 2, \dots, 1$, set*

$$Y_j = P_{c_j}(Y_{j+1}).$$

Then $D_S := Y_1$ is exactly the set of possible distances $|k - s_n|$ for $k \in K_S$. Consequently,

$$K_S = (s_n - D_S) \cup (s_n + D_S), \quad |K_S| = 2|D_S| - \mathbf{1}_{\{0 \in D_S\}}.$$

Proof. After appending k , follow the new right edge by setting

$$r_0 = k, \quad r_{j+1} = |r_j - c_j| \quad (0 \leq j < n).$$

All left-edge conditions inherited from S already equal 1, so the appended sequence is Gilbreath if and only if $r_n = 1$. For nonnegative c, y , the nonnegative solutions of $|x - c| = y$ are $c + y$, together with $c - y$ exactly when $y \leq c$. Reversing from the terminal value $r_n = 1$ through $c_{n-1}, \dots, c_1$ gives precisely the set D_S of possible $r_1 = |k - s_n|$. Every positive distance produces the two integers $s_n - d$ and $s_n + d$, while distance zero produces only s_n , which proves the cardinality formula. $\square$

Lemma 3 (complete generation). *The children of $S \in \mathcal{G}_n$ in $\mathcal{G}_{n+1}$ are exactly*

$$(S, s_n + d) \quad (d \in D_S, d > 0).$$

Starting at $(2, 3)$ and applying this rule therefore visits each member of $\mathcal{G}_m$ exactly once for every $m \geq 2$.

Proof. By Lemma 2, $s_n + d \in K_S$ for every $d \in D_S$; the condition $d > 0$ is exactly strict increase at the new coordinate. Conversely, deleting the last coordinate of a member of $\mathcal{G}_{n+1}$ gives its unique prefix $S \in \mathcal{G}_n$, and its positive last gap belongs to D_S . Since $\mathcal{G}_2 = \{(2, 3)\}$, induction gives both completeness and absence of duplicates. $\square$

3 Exhaustive computation

The primary C++17 program implements Lemmas 2 and 3. At each node it constructs D_S , computes the complete two-sided set K_S , updates the level count and extremum, retains every tie, and then recurses on every positive $d \in D_S$. Thus it does not assume the proposed upper bound and does not stop on finding U_{11} .

All set operations are exact. Intermediate distance sets are stored in increasing order. If Y is increasing, the values $c - y$ with $y \leq c$ are emitted while traversing Y backwards, and the values $c + y$ are emitted while traversing it forwards. The two branches can coincide only at c , when $y = 0$, and that single duplicate is removed. No floating-point operation is used.

The fixed storage and integer types can be justified before the run. There are $n - 1$ binary reverse steps, so $|D_S| \leq 2^{n-1}$ through level 11. If $M = s_n$, each positive-row entry of the difference triangle is at most $M - 2$, whence

$$d \leq 1 + \sum_{j=1}^{n-1} c_j \leq 1 + (n - 1)(M - 2).$$

A generated last coordinate is therefore less than nM . Beginning with $M = 3$ at level 2 gives $M < 3 \cdot 2 \cdot 3 \cdots 10 = 10,886,400$ through level 11; the resulting complete extension ranges also lie safely inside signed 32-bit range. Subtractions used for absolute differences are performed in signed 64-bit temporaries.

The exhaustive output was as follows.

n	$ \mathcal{G}_n $	$\max_{S \in \mathcal{G}_n} K_S $	number of maximizers
2	1	3	1
3	1	5	1
4	2	9	1
5	6	17	1
6	27	33	1
7	180	65	1
8	1,786	129	1
9	26,094	257	1
10	559,127	513	1
11	17,535,396	1025	1

At level 11 the sole retained maximizer was U_{11} . Its right diagonal is

$$(1025, 512, 256, 128, 64, 32, 16, 8, 4, 2, 1).$$

The reverse recursion gives

$$D_{U_{11}} = \{0, 2, 4, \dots, 1024\}, \quad K_{U_{11}} = \{1, 3, 5, \dots, 2049\},$$

so its width is directly checked to be 1025.

Three replay checks guard distinct failure modes. First, an assertion-enabled rebuild repeated the primary run and its complete records. Second, a separate full-depth C++ implementation used signed 64-bit values, generated preimages without the ordered merge, and applied a general sort-and-unique operation; it independently reproduced every row of the table and the sole level-11 maximizer. Third, a Python implementation rebuilt full difference triangles and used mathematical sets to reproduce all complete records through $n = 10$; at level 11 it forward-scanned the proved

finite interval for U_{11} and recovered exactly the odd integers from 1 to 2049. These are deterministic exact checks, not probabilistic tests.

For broad methodological comparison only, exact finite enumeration with separately replayable checks also appears in a different enumerative setting in [2]. That work concerns permutation cycle lengths and supplies no result about Gilbreath sequences.

Proof of Theorem 1. Lemma 3 proves that the traversal covers all and only the 17,535,396 members of $\mathcal{G}_{11}$, without duplication. Lemma 2 proves that the width computed at each visited node is the cardinality of its complete extension set over all integers. Exact enumeration gives maximum 1025, retains one equality case, and identifies that case as U_{11} . The independent full-depth implementation reproduces the same count, maximum, and equality classification. Hence the asserted maximum and the uniqueness statement follow. $\square$

4 Scope, reproducibility, and disclosure

The conclusion is confined to $n = 11$. It neither proves nor disproves the arbitrary- n conjecture. A targeted primary-literature check was performed through September 7, 2026; failure to locate another level-11 classification is not a proof of priority.

The accompanying reproducibility material consists of the primary C++ enumerator, machine-readable level and maximizer certificates, and the structurally separate Python verifier. The proof depends only on exact integer operations and the complete-generation argument above. A separate audit also rebuilt and ran a second full-depth implementation before accepting the finite claim.

Generative-AI systems assisted with mathematical exploration, implementation and review of verification code, proof auditing, and manuscript drafting. The theorem is not inferred from model output: its support is the explicit reverse recursion, the completeness induction, exact exhaustive enumeration, and independently implemented replay. The author takes responsibility for the statement and presentation.

References

- [1] Leila Muney. Holes in valid-extension sets of finite gilbreath sequences. arXiv:2606.23721v2, 2026. <https://arxiv.org/abs/2606.23721>.
- [2] Zijian Zeng. The distinct-cycle-length probability is strictly decreasing after 30. SSRN working paper 7380519, 2026. <https://doi.org/10.2139/ssrn.7380519>.